

PART A — QUESTIONS

AF11-001

Question:

Solve: $13x - 10 = 68$

- A) 5
- B) 6
- C) 7
- D) 8

AF11-002

Question:

If $11n + 6 = 83$, what is n ?

- A) 5
- B) 6
- C) 7
- D) 8

AF11-003

Question:

Simplify: $3(7x - 4) - 6x$

- A) $15x - 12$
- B) $21x - 12$
- C) $15x - 4$
- D) $27x - 12$

AF11-004

Question:

A company charges \$18 for each item plus a \$45 delivery fee. Which expression represents the total cost for i items?

- A) $18i + 45$
- B) $45i + 18$
- C) $63i$
- D) $45 - 18i$

AF11-005

Question:

If $f(x) = 8x - 15$, what is $f(7)$?

- A) 41
- B) 49
- C) 56
- D) 71

AF11-006

Question:

Which ordered pair satisfies the equation $y = 6x + 2$?

- A) (1, 7)
- B) (2, 14)
- C) (3, 20)
- D) (4, 25)

AF11-007

Question:

Solve: $12(x - 4) = 72$

- A) 8
- B) 9
- C) 10
- D) 11

AF11-008

Question:

A pattern begins: 6, 16, 26, 36, ... What is the next number?

- A) 44
- B) 45
- C) 46
- D) 48

AF11-009

Question:

Solve: $x/18 = 4$

- A) 22
- B) 54
- C) 72
- D) 84

AF11-010

Question:

A line has a slope of -8 and crosses the y-axis at 3. Which equation represents the line?

- A) $y = 3x - 8$
- B) $y = -8x + 3$
- C) $y = 8x + 3$
- D) $y = -3x + 8$

AF11-011

Question:

If 9 boxes contain 153 parts, how many parts are in 6 boxes at the same rate?

- A) 90
- B) 96
- C) 102
- D) 108

AF11-012

Question:

Solve: $72 - 10x = 22$

- A) 4
- B) 5
- C) 6
- D) 7

AF11-013

Question:

Which expression is equivalent to $35m - 16m + 7$?

- A) $19m + 7$
- B) $51m + 7$
- C) $19m - 7$
- D) $35m - 9$

AF11-014

Question:

A storage bin starts with 500 clips. Workers remove 35 clips each hour. Which equation gives the number of clips C left after h hours?

- A) $C = 500h - 35$
- B) $C = 35h + 500$
- C) $C = 500 - 35h$
- D) $C = 35h - 500$

AF11-015

Question:

If $y = -10x + 85$, what is y when $x = 7$?

- A) 10
- B) 15
- C) 70
- D) 155

AF11-016

Question:

Solve: $16x - 12 = 10x + 42$

- A) 7
- B) 8
- C) 9
- D) 10

AF11-017

Question:

Which value of x makes $9x - 8$ less than 55?

- A) 6
- B) 7
- C) 8
- D) 9

AF11-018

Question:

A table shows a linear relationship.

x : 0, 1, 2, 3

y : -4, 7, 18, 29

What is the rule?

- A) $y = 11x - 4$
- B) $y = 4x + 11$
- C) $y = -4x + 11$
- D) $y = 11x + 4$

AF11-019

Question:

Solve: $8(x + 3) = 5x + 48$

- A) 6
- B) 7
- C) 8
- D) 9

AF11-020

Question:

If $a = -5$ and $b = 9$, what is $3a + 2b$?

- A) -33
- B) -3
- C) 3
- D) 33

AF11-021

Question:

Which point is located in Quadrant II?

- A) (7, 5)
- B) (-7, 5)
- C) (-7, -5)
- D) (7, -5)

AF11-022

Question:

A worker sorts 99 pieces in 9 minutes. At the same rate, how many pieces can the worker sort in 5 minutes?

- A) 45
- B) 50
- C) 55
- D) 60

AF11-023

Question:

Solve for t : $R = 14t - 28$

- A) $t = R + 28$
- B) $t = (R + 28)/14$
- C) $t = 14(R + 28)$
- D) $t = R/14 - 28$

AF11-024

Question:

What is the slope of a line that goes through (3, 9) and (9, 45)?

- A) 4
- B) 5
- C) 6
- D) 36

AF11-025

Question:

Simplify: $25x - 7 - 16x + 19$

- A) $9x + 12$
- B) $41x + 12$
- C) $9x - 26$
- D) $41x - 26$

AF11-026

Question:

If $g(x) = x^2 - 6x$, what is $g(10)$?

- A) 30
- B) 40
- C) 60
- D) 100

AF11-027

Question:

A sequence follows the rule “multiply by 3, then subtract 7.” If the first term is 9, what is the second term?

- A) 20
- B) 22
- C) 27
- D) 34

AF11-028

Question:

Solve: $11(x - 2) + 6 = 72$

- A) 6
- B) 7
- C) 8
- D) 9

AF11-029

Question:

A line crosses the y-axis at 12 and falls 5 units for every 1 unit to the right. Which equation represents the line?

- A) $y = 5x + 12$
- B) $y = -5x + 12$
- C) $y = 12x - 5$
- D) $y = -12x + 5$

AF11-030

Question:

If $\frac{13}{16} = \frac{x}{48}$, what is x ?

- A) 26
- B) 32
- C) 36
- D) 39

AF11-031

Question:

Solve: $-13x + 26 = -78$

- A) -8
- B) -4
- C) 4
- D) 8

AF11-032

Question:

A warehouse starts with 1,000 tags. Each order uses 50 tags. Which expression gives the number of tags left after o orders?

- A) $1,000 + 50o$
- B) $1,000 - 50o$
- C) $50o - 1,000$
- D) $1,000 \div 50o$

AF11-033

Question:

Which equation has $x = 16$ as its solution?

- A) $2x + 9 = 39$
- B) $3x - 11 = 37$
- C) $x/4 + 6 = 11$
- D) $5x - 12 = 70$

PART B — ANSWER KEY + EXPLANATIONS

AF11-001 — Correct: B — Explanation:

Solve $13x - 10 = 68$. Add 10 to both sides: $13x = 78$. Divide by 13: $x = 6$. The most common mistake is subtracting 10 instead of adding 10.

AF11-002 — Correct: C — Explanation:

Start with $11n + 6 = 83$. Subtract 6 from both sides: $11n = 77$. Divide by 11: $n = 7$. A common mistake is stopping at 77 and forgetting to divide by 11.

AF11-003 — Correct: A — Explanation:

Distribute first: $3(7x - 4) = 21x - 12$. Then subtract $6x$: $21x - 6x - 12 = 15x - 12$. A common mistake is forgetting to multiply -4 by 3.

AF11-004 — Correct: A — Explanation:

The company charges \$18 for each item, so that part is $18i$. The \$45 delivery fee is added one time. The total cost is $18i + 45$. A common mistake is treating the delivery fee as the per-item charge.

AF11-005 — Correct: A — Explanation:

Substitute 7 for x : $f(7) = 8(7) - 15 = 56 - 15 = 41$. A common mistake is adding 15 instead of subtracting it.

AF11-006 — Correct: C — Explanation:

Test the choices in $y = 6x + 2$. For $(3, 20)$, $6(3) + 2 = 18 + 2 = 20$, which matches the y -value. A common mistake is choosing a point that is close but does not make the equation true.

AF11-007 — Correct: C — Explanation:

Solve $12(x - 4) = 72$. Divide both sides by 12: $x - 4 = 6$. Add 4: $x = 10$. A common mistake is adding 4 before undoing the multiplication by 12.

AF11-008 — Correct: C — Explanation:

The pattern increases by 10 each time: 6, 16, 26, 36. Add 10 to 36: 46. A common mistake is assuming the next number is 45 because the pattern looks close to adding 9.

AF11-009 — Correct: C — Explanation:

$x/18 = 4$ means x divided by 18 equals 4. Multiply both sides by 18: $x = 72$. A common mistake is adding 18 and 4 instead of multiplying.

AF11-010 — Correct: B — Explanation:

Slope-intercept form is $y = mx + b$. The slope is -8 and the y-intercept is 3, so the equation is $y = -8x + 3$. A common mistake is switching the slope and y-intercept.

AF11-011 — Correct: C — Explanation:

Find the unit rate: $153 \text{ parts} \div 9 \text{ boxes} = 17 \text{ parts per box}$. For 6 boxes: $17 \times 6 = 102$. A common mistake is multiplying 153 by 6 without finding the rate first.

AF11-012 — Correct: B — Explanation:

Solve $72 - 10x = 22$. Subtract 72 from both sides: $-10x = -50$. Divide by -10: $x = 5$. A common mistake is losing the negative sign after subtracting 72.

AF11-013 — Correct: A — Explanation:

Combine like terms: $35m - 16m = 19m$. The constant 7 stays the same. The expression becomes $19m + 7$. A common mistake is combining the constant with the variable terms.

AF11-014 — Correct: C — Explanation:

The bin starts with 500 clips. Workers remove 35 clips each hour, so after h hours, $35h$ clips are removed. The number left is $C = 500 - 35h$. A common mistake is adding $35h$ even though clips are being removed.

AF11-015 — Correct: B — Explanation:

Substitute $x = 7$ into $y = -10x + 85$: $y = -10(7) + 85 = -70 + 85 = 15$. A common mistake is treating $-10(7)$ as positive 70.

AF11-016 — Correct: C — Explanation:

Solve $16x - 12 = 10x + 42$. Subtract $10x$: $6x - 12 = 42$. Add 12: $6x = 54$. Divide by 6: $x = 9$. A common mistake is adding 12 to only one side.

AF11-017 — Correct: A — Explanation:

Solve $9x - 8 < 55$. Add 8: $9x < 63$. Divide by 9: $x < 7$. The only listed value less than 7 is 6. A common mistake is choosing 7, but $9(7) - 8 = 55$, not less than 55.

AF11-018 — Correct: A — Explanation:

The y-values increase by 11 each time x increases by 1, so the slope is 11. When $x = 0$, $y = -4$, so the y-intercept is -4. The rule is $y = 11x - 4$. A common mistake is using +4 instead of -4.

AF11-019 — Correct: C — Explanation:

Distribute: $8x + 24 = 5x + 48$. Subtract $5x$: $3x + 24 = 48$. Subtract 24: $3x = 24$. Divide by 3: $x = 8$. A common mistake is forgetting to multiply 3 by 8.

AF11-020 — Correct: C — Explanation:

Substitute $a = -5$ and $b = 9$: $3a + 2b = 3(-5) + 2(9) = -15 + 18 = 3$. A common mistake is treating $3(-5)$ as positive 15.

AF11-021 — Correct: B — Explanation:

Quadrant II has a negative x-coordinate and a positive y-coordinate. The point $(-7, 5)$ fits this description. A common mistake is choosing $(7, -5)$, which is Quadrant IV.

AF11-022 — Correct: C — Explanation:

99 pieces in 9 minutes means $99 \div 9 = 11$ pieces per minute. In 5 minutes: $11 \times 5 = 55$. A common mistake is multiplying 99 by 5 without finding the rate first.

AF11-023 — Correct: B — Explanation:

Start with $R = 14t - 28$. Add 28 to both sides: $R + 28 = 14t$. Divide by 14: $t = (R + 28)/14$. A common mistake is dividing R by 14 first and then adding 28.

AF11-024 — Correct: C — Explanation:

Slope equals change in y divided by change in x . From $(3, 9)$ to $(9, 45)$, y increases by 36 and x increases by 6. The slope is $36 \div 6 = 6$. A common mistake is using 36 as the slope without dividing by the change in x .

AF11-025 — Correct: A — Explanation:

Combine like terms: $25x - 16x = 9x$, and $-7 + 19 = 12$. The simplified expression is $9x + 12$. A common mistake is writing $9x - 12$ because the positive 19 is missed.

AF11-026 — Correct: B — Explanation:

Substitute 10 for x : $g(10) = 10^2 - 6(10) = 100 - 60 = 40$. A common mistake is calculating 10^2 as 20 instead of 100.

AF11-027 — Correct: A — Explanation:

Start with 9. Multiply by 3: 27. Then subtract 7: 20. A common mistake is subtracting 7 first and then multiplying.

AF11-028 — Correct: C — Explanation:

Solve $11(x - 2) + 6 = 72$. Distribute: $11x - 22 + 6 = 72$. Combine: $11x - 16 = 72$. Add 16: $11x = 88$. Divide by 11: $x = 8$. A common mistake is forgetting to combine -22 and $+6$.

AF11-029 — Correct: B — Explanation:

“Falls 5 units for every 1 unit to the right” means the slope is -5 . The line crosses the y -axis at 12, so the equation is $y = -5x + 12$. A common mistake is using a positive slope because the number 5 is given.

AF11-030 — Correct: D — Explanation:

Use the proportion $13/16 = x/48$. Since $16 \times 3 = 48$, multiply 13 by 3: $x = 39$. A common mistake is adding 32 to the numerator because 16 becomes 48.

AF11-031 — Correct: D — Explanation:

Solve $-13x + 26 = -78$. Subtract 26 from both sides: $-13x = -104$. Divide by -13 : $x = 8$. A common mistake is giving -8 because both sides contain negative numbers.

AF11-032 — Correct: B — Explanation:

The warehouse starts with 1,000 tags. Each order uses 50 tags, so o orders use $50o$ tags. The number left is $1,000 - 50o$. A common mistake is adding $50o$ even though tags are being used.

AF11-033 — Correct: B — Explanation:

Test $x = 16$. In choice B, $3(16) - 11 = 48 - 11 = 37$, so the equation is true. A common mistake is choosing a nearby equation without checking both sides exactly.